

16 - Module 1 Complete Verification

The honest status of every gate, and how to close the ones that are open.

What is this?

A checklist of everything Module 1 must satisfy, with the current state of each and the exact command that proves it.

IMPORTANT

Nothing here is marked COMPLETE without evidence
"The code exists" is not evidence. "This command was run and produced this output" is evidence. Several gates below are marked **BLOCKED**, meaning the work is done but could not be verified in the environment where it was written. You must close those on your own machine.

Status key

Word	Meaning
COMPLETE	Built, and verified by a command that was actually executed
PARTIAL	Built, but part of the required scope is missing
BLOCKED	Built, but could not be verified here due to the environment
NOT VERIFIED	Built, but nobody has checked it yet

The status table

Component	Status	Evidence
Backend source	COMPLETE	88 Java files; <code>static_check.py</code> passes
PostgreSQL migration	COMPLETE	No MySQL syntax in any executable line; checker enforces it
Flyway migrations	NOT VERIFIED	2 files written; never executed against a live database
Seed data	PARTIAL	Written; BCrypt hashes verified against plaintext; migration never run
Backend tests	BLOCKED	11 files, 149 methods written; never executed
Testcontainers	BLOCKED	<code>PostgreSQLContainer("postgres:16-alpine")</code> wired; Docker unavailable
Swagger	NOT VERIFIED	Annotations and config written; page never opened

Component	Status	Evidence
Postman	COMPLETE	54 requests, all 25 endpoints covered; JSON validated
Frontend	COMPLETE	71 files, 9 pages; <code>npm run build</code> exit 0
Frontend/API integration	NOT VERIFIED	Services match controllers; never run against a live backend
Documentation	PARTIAL	6 of 18 planned PDFs written
Architecture diagrams	NOT STARTED	Planned for document 18
Static checks	COMPLETE	<code>python3 registry/static_check.py</code> passes
Maven verify	BLOCKED	Maven Central unreachable in the authoring environment
Full integration tests	BLOCKED	Requires Docker
Module 1 overall	NOT COMPLETE	5 gates blocked, 2 partial

Why some gates are blocked

IMPORTANT

BUG-ENV-001, recorded in the bug registry

The environment this code was written in cannot reach Maven Central and cannot run Docker. Therefore:

- `mvn clean verify` has never run - the backend has **never been compiled**
- The 149 tests have never executed
- Flyway has never applied a migration to a real database
- Swagger has never been opened

Expect to fix a small number of import-level errors on your first compile.

That is normal for code that has been statically checked but not built. The static checker verifies package declarations, brace and parenthesis balance, entity-to-migration column agreement, interface-to-implementation method coverage, permission constants, and seed-to-schema column agreement - but it cannot resolve a Java type.

Closing the blocked gates

Run these in order on a machine with Docker and internet access.

Gate 1 - the backend compiles

RUN THIS

From the project root

```
cd backend
./mvnw clean compile
```

WHAT YOU SHOULD SEE

Expected

```
[INFO] BUILD SUCCESS
```

IF IT FAILS

If it fails

Read the first error only; later ones are usually consequences. The likely causes are a missing import or a method signature that drifted. Fix, re-run. Record anything you fix in `project-knowledge/BUG_REGISTRY.md`.

Gate 2 - the tests pass

RUN THIS

Docker must be running

```
docker info > /dev/null && echo "docker ok"
cd backend
./mvnw clean verify
```

WHAT YOU SHOULD SEE

Expected

```
Tests run: 149, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
```

The first run is slow - Testcontainers downloads postgres:16-alpine.

VERIFY

What passing actually proves

- Real PostgreSQL, not H2, so CHECK constraints and functional indexes are genuinely exercised
- Flyway ran V1 and V900 successfully against a real database
- The seeded BCrypt hashes really do authenticate
- Enumeration protection holds: wrong-password and unknown-account responses are byte-identical
- Refresh token reuse revokes every session
- The last active owner cannot be removed

Gate 3 - Flyway and seed data on your own database

RUN THIS

```
docker compose up -d
cd backend && ./mvnw spring-boot:run
```

RUN THIS

In another terminal

```
docker exec -it hardware-erp-postgres psql -U hardware_erp -d hardware_erp \
-c "SELECT version, description, success FROM flyway_schema_history;"
docker exec -it hardware-erp-postgres psql -U hardware_erp -d hardware_erp \
-c "SELECT count(*) AS users FROM app_user;"
```

WHAT YOU SHOULD SEE

Expected

Versions 1 and 900, both success = t, and 12 users.

Gate 4 - Swagger

RUN THIS

With the backend running, open in a browser

```
http://localhost:8080/api/swagger-ui.html
```

WHAT YOU SHOULD SEE

Expected

A page titled "Hardware ERP API", with five tag groups: Authentication, Users, Roles, Permissions and Security Audit Log. Expand

POST /v1/auth/login, click **Try it out**, send

`{"identifier": "9876543210", "password": "Owner@2026"}` and get a 200 with a token. Copy the accessToken value, click **Authorize** at the top right, paste it, and protected endpoints then work.

Gate 5 - Postman

Follow document 09. The Collection Runner should report all 54 requests green.

Gate 6 - the frontend against the live backend

RUN THIS

```
cd frontend
npm install
npm run dev
```

Then open `http://localhost:5173` and sign in as `9876543210 / Owner@2026`.

VERIFY

Walk through every page

1. **Login** - sign in as OWNER
2. **Profile** - your name, role and permissions appear; the Sessions tab lists this device
3. **Users** - 11 users listed, the soft-deleted one absent; search for "Karthik" narrows it to one
4. Create a user, edit it, deactivate it; confirm it leaves the list
5. **Roles** - five roles; open STAFF and confirm PRODUCT_VIEW_COST is unticked
6. **Permissions** - 31 permissions grouped by module
7. **Security log** - shows the LOGIN_SUCCESS your sign-in just created
8. Sign out, then sign in as STAFF (9843012345 / Staff@2026). The Users, Roles and Security log links must be **absent** from the sidebar
9. With STAFF signed in, type /users in the address bar directly. You should get an access message, not the user list

WHY

Why step 9 matters

Steps 1 to 8 test the happy path. Step 9 tests that hiding a link is not the only thing stopping access. Even if the guard were removed, every API call the page makes would return 403, because the backend checks the permission independently.

The already-verified gates

These were genuinely executed. The commands and their output:

Static consistency check

RUN THIS

```
python3 registry/static_check.py
```

WHAT YOU SHOULD SEE

Actual output

```
=====
STATIC CONSISTENCY CHECK
88 main sources, 11 test sources, 7 tables
=====

PASS - no structural inconsistencies
NOTE: this is not a compile.
```

Frontend build

RUN THIS

```
cd frontend && npm run build
```

WHAT YOU SHOULD SEE

Actual output

```
vite v6.4.3 building for production...
1794 modules transformed.
dist/assets/vendor-forms-*.js      87.82 kB
dist/assets/vendor-ui-*.js        114.92 kB
dist/assets/vendor-react-*.js     165.80 kB
dist/assets/index-*.js            193.54 kB
built in 6.72s
```

Postman endpoint coverage

WHAT YOU SHOULD SEE

Actual output

```
BACKEND ENDPOINTS: 25
COVERED BY POSTMAN: 25
MISSING: none - full coverage
```

What is genuinely not built yet

IMPORTANT

Remaining work, stated plainly

- **12 of the 18 planned PDFs.** Written: 01, 02, 03, 04, 09, 16.
Not written: 05, 06, 07, 08, 10, 11, 12, 13, 14, 15, 17, 18.
- **Architecture diagrams** (document 18) - none produced yet
- **Frontend unit tests** - src/modules/auth/tests/ is an empty folder
- **Swagger has never been opened**, so the annotations are unproven

Sign-off

VERIFY

Module 1 may be called COMPLETE only when all of these are true

☐ mvn clean compile	BUILD SUCCESS
☐ mvn clean verify	149 tests, 0 failures
☐ docker compose up -d	container healthy
☐ Flyway	versions 1 and 900, success = t
☐ Seed data	12 users present
☐ Swagger	loads, Authorize works, endpoints respond
☐ Postman	54 requests green
☐ Frontend build	exit 0
☐ Frontend to backend	all 9 pages work against the live API
☐ STAFF restriction	cannot reach /users even by typing the URL
☐ static_check.py	PASS
☐ Documentation	all 18 PDFs written
☐ Diagrams	produced

Until every line is ticked, the status stays **NOT COMPLETE**. Module 2 does not start before then.